# Derivation of the Clopper-Pearson confidence interval and proof of its coverage

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This note provides a mathematical derivation of the Clopper-Pearson confidence interval, introduced by C. J. Clopper and E. S. Pearson [1], and proof of its coverage level. The results presented here are classical and are not original to this note. They are derived independently to ensure a thorough understanding of the underlying mathematics, to present the arguments in a form suited to this project, and to obtain conclusions that are directly useful for the implementation in Problab[1]. The derivation assumes $K_p \sim Bin(n,p)$ and a significance level $\alpha \in (0,1)$, corresponding to a target coverage level of $1-\alpha$. In Lemma 1 and Lemma 2 the lower and upper Clopper-Pearson endpoints, $p_L(k)$ and $p_U(k)$, are derived. In Lemma 3 and Lemma 4, it is proven that $P(p < p_L(K_p)) < \alpha/2$ and $P(p > p_U(K_p)) < \alpha/2$. Lastly, in Corollary 5 it is shown that

$$P(p_L(K_p) \leq p \leq p_U(K_p)) \geq 1-\alpha.$$

Practically, these results enable Problab to attach a $1-\alpha$ confidence interval to probabilities estimated from samples, providing a measure of the uncertainty associated with probability estimates involving random variables.

**Lemma 1.** *Let $n \in \mathbb{Z}_{>0}$, $\alpha \in (0,1)$, $p \in [0,1]$, and $K_p \sim Bin(n,p)$. For $k \in \{1,\ldots,n\}$,*

$$P(K_p \geq k) = F_{Beta(k,n-k+1)}(p).$$

*Therefore,*

$$P(K_p \geq k) = \alpha/2$$

*if and only if*

$$p = F^{-1}_{Beta(k,n-k+1)}(\alpha/2).$$

*Proof.* It is commonly known, that

$$P(K_p \geq k) = \sum_{j=k}^{n} \binom{n}{j} p^j (1-p)^{n-j}.$$

If you will indulge me for a moment, consider what happens when differentiate this expression with respect to $p \in (0,1)$:

$$\frac{d}{dp} \sum_{j=k}^{n} \binom{n}{j} p^j (1-p)^{n-j} = \sum_{j=k}^{n} \binom{n}{j} \frac{d}{dp} \left(p^j (1-p)^{n-j}\right)$$

[1] `https://github.com/FrederikDanielsenDevelopment/problab`

$$= \sum_{j=k}^{n} \binom{n}{j} \left( jp^{j-1}(1-p)^{n-j} - p^j(n-j)(1-p)^{n-j-1} \right)$$

$$= \sum_{j=k}^{n} \binom{n}{j} jp^{j-1}(1-p)^{n-j} - \sum_{j=k}^{n} \binom{n}{j} p^j(n-j)(1-p)^{n-j-1}$$

Using the identities

$$j\binom{n}{j} = n\binom{n-1}{j-1}$$

and

$$(n-j)\binom{n}{j} = n\binom{n-1}{j},$$

together with the standard conventions

$$\binom{m}{r} = 0 \,\forall\, r > m$$

and

$$\sum_{j=a}^{b} x_j = 0 \quad \text{if } a > b,$$

we get

$$\sum_{j=k}^{n} \binom{n}{j} jp^{j-1}(1-p)^{n-j} - \sum_{j=k}^{n} \binom{n}{j} p^j(n-j)(1-p)^{n-j-1}$$

$$= \sum_{j=k}^{n} n\binom{n-1}{j-1} p^{j-1}(1-p)^{n-j} - \sum_{j=k}^{n} p^j n\binom{n-1}{j}(1-p)^{n-j-1}$$

$$= \sum_{j=k-1}^{n-1} n\binom{n-1}{j} p^j(1-p)^{n-j-1} - \left( \sum_{j=k}^{n-1} p^j n\binom{n-1}{j}(1-p)^{n-j-1} + p^n n\binom{n-1}{n}(1-p)^{-1} \right)$$

$$= \sum_{j=k-1}^{n-1} n\binom{n-1}{j} p^j(1-p)^{n-j-1} - \sum_{j=k}^{n-1} p^j n\binom{n-1}{j}(1-p)^{n-j-1}$$

$$= n\binom{n-1}{k-1} p^{k-1}(1-p)^{n-k} + \sum_{j=k}^{n-1} n\binom{n-1}{j} p^j(1-p)^{n-j-1} - \sum_{j=k}^{n-1} p^j n\binom{n-1}{j}(1-p)^{n-j-1}$$

$$= n\binom{n-1}{k-1} p^{k-1}(1-p)^{n-k}$$

$$= \frac{n!}{(k-1)!(n-k)!} p^{k-1}(1-p)^{n-k}$$

The Beta probability density function is given by

$$f_{Beta(a,b)}(x) = \frac{x^{a-1}(1-x)^{b-1}}{B(a,b)},$$

where

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

and $\Gamma$ denotes the Gamma function. For any positive integer $n$ it holds that $\Gamma(n) = (n-1)!$. Notice that if $a = k$ and $b = n - k + 1$ then

$$\begin{aligned} f_{Beta(k,n-k+1)}(p) &= \frac{p^{k-1}(1-p)^{n-k}}{B(k,n-k+1)} = \frac{\Gamma(n+1)p^{k-1}(1-p)^{n-k}}{\Gamma(k)\Gamma(n-k+1)} \\ &= \frac{n!}{(k-1)!(n-k)!}p^{k-1}(1-p)^{n-k} = \frac{d}{dp}P(K_p \geq k) \end{aligned}$$

Hence,

$$P(K_p \geq k) - P(K_0 \geq k) = \int_0^p \frac{d}{dt}P(K_t \geq k)dt = \int_0^p f_{Beta(k,n-k+1)}(t)dt.$$

When $p = 0$ the binomial random variable $K$ is equal to 0 with probability 1. Therefore, for $k \in \{1, \dots, n\}$, the event $K \geq k$ is impossible and hence $P(K_0 \geq k) = 0$. Thus

$$P(K_p \geq k) = \int_0^p f_{Beta(k,n-k+1)}(t)dt = F_{Beta(k,n-k+1)}(p).$$

Note that this equality also holds at $p = 0$ and $p = 1$ by continuity. Since $F_{Beta(k,n-k+1)}$ is strictly increasing, imposing the constraint $P(K_p \geq k) = \alpha/2$ gives

$$P(K_p \geq k) = \alpha/2 \Leftrightarrow p = F^{-1}_{Beta(k,n-k+1)}(\alpha/2).$$

$\square$

**Lemma 2.** *Let* $n \in \mathbb{Z}_{>0}$, $\alpha \in (0,1)$, $p \in [0,1]$, *and* $K_p \sim Bin(n,p)$. *For* $k \in \{0, \dots, n-1\}$,

$$P(K_p \leq k) = 1 - F_{Beta(k+1,n-k)}(p).$$

*Therefore,*

$$P(K_p \leq k) = \alpha/2$$

*if and only if*

$$p = F^{-1}_{Beta(k+1,n-k)}(1-\alpha/2).$$

*Proof.* It is commonly known, that

$$P(K_p \leq k) = \sum_{j=0}^{k} \binom{n}{j} p^j (1-p)^{n-j}.$$

If you will indulge me for a moment, consider what happens when differentiate this expression with respect to $p \in (0,1)$:

$$\frac{d}{dp}\sum_{j=0}^{k}\binom{n}{j}p^j(1-p)^{n-j} = \sum_{j=0}^{k}\binom{n}{j}\frac{d}{dp}\left(p^j(1-p)^{n-j}\right)$$

$$= \sum_{j=0}^{k}\binom{n}{j}\left(jp^{j-1}(1-p)^{n-j} - p^j(n-j)(1-p)^{n-j-1}\right)$$

$$= \sum_{j=0}^{k}\binom{n}{j}jp^{j-1}(1-p)^{n-j} - \sum_{j=0}^{k}\binom{n}{j}p^j(n-j)(1-p)^{n-j-1}$$

Using the identities

$$j\binom{n}{j} = n\binom{n-1}{j-1}$$

and

$$(n-j)\binom{n}{j} = n\binom{n-1}{j},$$

together with the standard conventions

$$\binom{m}{r} = 0 \; \forall\, r > m$$

and

$$\sum_{j=a}^{b} x_j = 0 \quad \text{if } a > b,$$

we get

$$\sum_{j=0}^{k}\binom{n}{j}jp^{j-1}(1-p)^{n-j} - \sum_{j=0}^{k}\binom{n}{j}p^j(n-j)(1-p)^{n-j-1}$$

$$= \sum_{j=1}^{k}n\binom{n-1}{j-1}p^{j-1}(1-p)^{n-j} - \sum_{j=0}^{k}p^j n\binom{n-1}{j}(1-p)^{n-j-1}$$

$$= \sum_{j=0}^{k-1}n\binom{n-1}{j}p^{j}(1-p)^{n-j-1} - \sum_{j=0}^{k}p^j n\binom{n-1}{j}(1-p)^{n-j-1}$$

$$= \sum_{j=0}^{k-1}n\binom{n-1}{j}p^{j}(1-p)^{n-j-1} - \left(\sum_{j=0}^{k-1}p^j n\binom{n-1}{j}(1-p)^{n-j-1} + p^k n\binom{n-1}{k}(1-p)^{n-k-1}\right)$$

$$= -n\binom{n-1}{k}p^k(1-p)^{n-k-1}$$

$$= -\frac{n!}{k!(n-k-1)!} p^k (1-p)^{n-k-1}$$

The Beta probability density function is given by

$$f_{Beta(a,b)}(x) = \frac{x^{a-1}(1-x)^{b-1}}{B(a,b)},$$

where

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

and $\Gamma$ denotes the Gamma function. For any positive integer $n$ it holds that $\Gamma(n) = (n-1)!$. Notice that if $a = k+1$ and $b = n-k$ then

$$\begin{aligned} f_{Beta(k+1,n-k)}(p) &= \frac{p^k(1-p)^{n-k-1}}{B(k+1,n-k)} = \frac{\Gamma(n+1)p^k(1-p)^{n-k-1}}{\Gamma(k+1)\Gamma(n-k)} \\ &= \frac{n!}{k!(n-k-1)!} p^k (1-p)^{n-k-1} = -\frac{d}{dp} P(K_p \leq k) \end{aligned}$$

Hence,

$$P(K_p \leq k) - P(K_0 \leq k) = \int_0^p \frac{d}{dt} P(K_t \leq k) dt = -\int_0^p f_{Beta(k+1,n-k)}(t) dt.$$

When $p = 0$ the binomial random variable $K$ is equal to 0 with probability 1. Therefore, for $k \in \{0, \ldots, n-1\}$, the event $K \leq k$ occurs with probability 1 and hence $P(K_0 \leq k) = 1$. Thus

$$P(K_p \leq k) = 1 - \int_0^p f_{Beta(k+1,n-k)}(t) dt = 1 - F_{Beta(k+1,n-k)}(p).$$

Note that this equality also holds at $p = 0$ and $p = 1$ by continuity. Since $F_{Beta(k+1,n-k)}$ is strictly increasing, imposing the constraint $P(K_p \leq k) = \alpha/2$ gives

$$P(K_p \leq k) = 1 - F_{Beta(k+1,n-k)}(p) = \alpha/2 \Leftrightarrow p = F^{-1}_{Beta(k+1,n-k)}(1 - \alpha/2).$$

$\square$

**Lemma 3.** *Let $n \in \mathbb{Z}_{>0}$, $\alpha \in (0,1)$, $p \in [0,1]$, $K_p \sim Bin(n,p)$, and*

$$p_L(k) := \begin{cases} 0, & k = 0, \\ F^{-1}_{Beta(k,n-k+1)}(\alpha/2), & k \in \{1, \ldots, n\}. \end{cases}$$

*Then*

$$P(p_L(K_p) > p) < \alpha/2.$$

*Proof.* As stated in Lemma 1, for $k \in \{1, \dots, n\}$,

$$P(K_p \geq k) = \alpha/2 \Leftrightarrow p = p_L(k).$$

Furthermore, increasing the probability of success, $p$, increases the likelihood of seeing $k$ or more successes. In other words, $P(K_p \geq k)$ is strictly increasing in $p$. This implies that

$$p < p_L(k) \Leftrightarrow P(K_p \geq k) < \alpha/2.$$

In the case that $(n, p)$ is so that $P(K_p \geq k) \not< \alpha/2$ for all $k \in \{1, \dots, n\}$, then necessarily $p \geq p_L(k)$ for all said $k$. Furthermore, since by definition $p_L(0) = 0 \leq p$ for all $p \in [0, 1]$ this would mean that

$$p \geq p_L(k), \ \forall\, k \in \{0, \dots, n\}.$$

Consequently, in this case, we have

$$P(p_L(K_p) > p) = 0 < \alpha/2.$$

Otherwise, inevitably $\exists\, k^* \in \{1, \dots, n\}$ such that

$$k^* = \min\{k \in \{1, \dots, n\} : \, P(K_p \geq k) < \alpha/2\}.$$

Since $P(K_p \geq k)$ decreases in $k$, for any $k \geq k^*$ the inequality $P(K_p \geq k) < \alpha/2$ holds. By the definition of $k^*$, it does not hold for any $k < k^*$. Therefore

$$P(K_p \geq k) < \alpha/2 \Leftrightarrow k \geq k^*, \quad \forall\, k \in \{1, \dots, n\}$$

so, for $k \in \{1, \dots, n\}$,

$$p < p_L(k) \Leftrightarrow k \geq k^*.$$

Note that since $p_L(0) = 0 \leq p$,

$$p < p_L(0)$$

is false. Furthermore, since $k^* \in \{1, \dots, n\}$,

$$0 \geq k^*$$

is also false. Therefore

$$p < p_L(k) \Leftrightarrow k \geq k^*, \quad \forall\, k \in \{0, \dots, n\}.$$

Thus, the following events are equal:

$$\{p < p_L(K_p)\} = \{K_p \geq k^*\}$$

which implies that

$$P(p_L(K_p) > p) = P(K_p \geq k^*) < \alpha/2.$$

$\square$

**Lemma 4.** *Let* $n \in \mathbb{Z}_{>0}$*,* $\alpha \in (0,1)$*,* $p \in [0,1]$*,* $K_p \sim Bin(n,p)$*, and*

$$p_U(k) := \begin{cases} F^{-1}_{Beta(k+1,n-k)}(1-\alpha/2), & k \in \{0,\dots,n-1\}, \\ 1, & k = n. \end{cases}$$

*Then*

$$P(p_U(K_p) < p) < \alpha/2.$$

*Proof.* As stated in Lemma 2, for $k \in \{0,\dots,n-1\}$,

$$P(K_p \leq k) = \alpha/2 \Leftrightarrow p = p_U(k).$$

Furthermore, increasing the probability of success, $p$, decreases the likelihood of seeing $k$ or less successes. In other words, $P(K_p \leq k)$ is strictly decreasing in $p$. This implies that

$$p > p_U(k) \Leftrightarrow P(K_p \leq k) < \alpha/2.$$

In the case that $(n,p)$ is so that $P(K_p \leq k) \not< \alpha/2$ for all $k \in \{0,\dots,n-1\}$, then necessarily $p \leq p_U(k)$ for all said $k$. Furthermore, since by definition $p_U(n) = 1 \geq p$ for all $p \in [0,1]$ this would mean that

$$p \leq p_U(k),\ \forall\, k \in \{0,\dots,n\}.$$

Consequently, in this case, we have

$$P(p_U(K_p) < p) = 0 < \alpha/2.$$

Otherwise, inevitably $\exists\, k^* \in \{0,\dots,n-1\}$ such that

$$k^* = \max\{k \in \{0,\dots,n-1\} : P(K_p \leq k) < \alpha/2\}.$$

Since $P(K_p \leq k)$ increases in $k$, for any $k \leq k^*$ the inequality $P(K_p \leq k) < \alpha/2$ holds. By the definition of $k^*$, it does not hold for any $k > k^*$. Therefore

$$P(K_p \leq k) < \alpha/2 \Leftrightarrow k \leq k^*, \quad \forall\, k \in \{0,\dots,n-1\},$$

so, for $k \in \{0,\dots,n-1\}$,

$$p > p_U(k) \Leftrightarrow k \leq k^*.$$

Note that since $p_U(n) = 1 \geq p$,

$$p > p_U(n)$$

is false. Furthermore, since $k^* \in \{0,\dots,n-1\}$,

$$n \leq k^*$$

is also false. Therefore

$$p > p_U(k) \Leftrightarrow k \leq k^*, \quad \forall\, k \in \{0,\dots,n\}.$$

Thus, the following events are equal:

$$\{p > p_U(K_p)\} = \{K_p \leq k^*\}$$

which implies that

$$P(p_U(K_p) < p) = P(K_p \leq k^*) < \alpha/2.$$

$\square$

**Corollary 5.** *Let $n \in \mathbb{Z}_{>0}$, $\alpha \in (0,1)$, $p \in [0,1]$, $K_p \sim Bin(n,p)$. Let $p_L$ and $p_U$ be as defined in Lemma 3 and Lemma 4. then*

$$P(p_L(K_p) \leq p \leq p_U(K_p)) > 1 - \alpha$$

*The interval $[p_L(K_p), p_U(K_p)]$ is commonly referred to as the Clopper-Pearson interval, as it was introduced by C. J. Clopper and E. S. Pearson in 1934 [1].*

*Proof.* In Lemma 3 and Lemma 4, it is proven that

$$\begin{aligned} P(p < p_L(K_p)) &< \alpha/2, \\ P(p > p_U(K_p)) &< \alpha/2. \end{aligned}$$

Since

$$\{p \notin [p_L(K_p),\, p_U(K_p)]\} = \{p < p_L(K_p)\} \cup \{p > p_U(K_p)\},$$

the union bound gives

$$\begin{aligned} P(p \notin [p_L(K_p),\, p_U(K_p)]) &\leq P(p < p_L(K_p)) + P(p > p_U(K_p)) \\ &< \alpha/2 + \alpha/2 \\ &= \alpha. \end{aligned}$$

Since

$$P(p \notin [p_L(K_p),\, p_U(K_p)]) + P(p \in [p_L(K_p),\, p_U(K_p)]) = 1$$

we can conclude that

$$P\left(p \in [p_L(K_p),\, p_U(K_p)]\right) > 1 - \alpha \Leftrightarrow P(p_L(K_p) \leq p \leq p_U(K_p)) > 1 - \alpha$$

$\square$

## References

[1] C. J. Clopper and E. S. Pearson, "The Use of Confidence or Fiducial Limits Illustrated in the Case of the Binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.